

INDIA'S EV, SPACE, CLIMATE FUTURE

HOW THE NATION'S PROGRESSING
THROUGH TECH IN THESE THREE KEY AREAS

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A ccording to Fortune Business Insights, India's EV journey is about to kick into gear big time, with the electric vehicle market projected to grow from \$3.21 billion in 2022 to \$113.99 billion by 2029. Let's see how the government, technological progress and startups are leading the EV charge...

India's EV road in 2023

Whether it's buying smartphones, laptops or electric vehicles, India is and will always remain a price-sensitive country when it comes to buying gadgets and gizmos of all sizes and shapes. Beyond just a lot of smart tech infusion, the electric scooter and car market in India has been blossoming because of key affordability and environmental sustainability concerns. Rising oil prices and higher operating costs of petrol, diesel and now even CNG-powered vehicles in the country are forcing consumers to look at EVs as a cost-effective alternative for their daily commute to work and beyond.

In terms of health and safety regulations, Phase 1 of the Indian government's EV battery safety standards came into effect on December 1, 2022. AIS 156 safety standard certification is for batteries meant for use in two and three-wheeler EVs, while AIS 038 (Rev 2) certifies battery safety for four-wheeler EVs and above. The deadline for enhanced R&D and design tweaks down to the battery packs marking Phase 2 of the government's safety regulation implementation is March 31, 2023. Hopefully, this will allay fears among the public and prevent EV batteries from catching fire.



Everyone from Ola Electric to Ather Energy, Tata Motors to Mahindra eMobility, all these companies are leading the charge on bringing everything from two-wheeled EVs to four-wheeled EVs and beyond to the Indian market, but until India ramps up domestic manufacturing of Li-ion batteries—a product that's hampered by lack of raw materials to spur local production and which is extremely costly to import (about 60% of an EV's cost is all batteries), the EV industry in India won't truly takeoff. While the likes of Ola, Reliance and Hyundai are helping scale battery manufacturing capacity, research is also

accelerating towards novel battery chemistry advancements, including new graphite-based batteries to power EV scooters.

For this year, we expect the EV instant-charging and charging stations ecosystem to widen and get denser in India's biggest metro cities and start to proliferate beyond